Apertus Sovereign Operations Bridge

Technical report | Hack Apertus Online | Track 2B | 13 October 2026

Team: SPINNENNETZ-DNA. Author and sole participant: Stefan Ambrosini, Switzerland. Public runtime baseline: fc39de8a574393a29bfc4e41568c458b086672b6. This update adds documentation and a prototype walkthrough; it does not change the runtime.

1. Summary

Business users need to distinguish a plausible answer from an answer supported by complete, consistent sources and appropriate authorization. This standalone prototype uses Apertus for semantic extraction and draft preparation, followed by deterministic checks. It returns VERIFIED, HUMAN_REVIEW or BLOCKED and retains an inspectable result. A model proposal cannot authorize an external action.

Five synthetic live cases using CSCS-hosted Apertus v1.5 70B reached their expected control states on 3 October 2026 [1]. A fresh browser walkthrough on 13 October demonstrates the same five deterministic control paths with explicitly simulated model responses. It is a recorded sequence of actual UI result views, not a new live inference benchmark.

2. Architecture

Synthetic request and source documents -> one Apertus call -> structured JSON -> independent deterministic control layer -> status, reasons and evidence -> browser or CLI readback.

The Python HTTP service accepts a scenario request. The model schema carries facts, source IDs, missing information, untrusted instructions, a proposed response and requested actions. Caller-observed protected actions are retained even when the model omits them. SHA-256 binds each result to its input. The public application has no email-send, payment, contract, printer or Office execution tool.

3. Use of Apertus

Model: swiss-ai/Apertus-v1.5-70B. Verified serving: authorized CSCS inference at https://api.inference.cscs.ch/v1. Role: language analysis and response preparation. No fine-tuning and no secondary judge model. ChatGPT/Codex assisted development; they are not runtime dependencies of the public prototype.

The client calls /chat/completions with temperature 0, max_tokens 1200 and JSON-object response format. It parses complete JSON objects, including responses wrapped in prose or code fences. Prompts, schema, fixtures and controls are in track_2b/src/app/. Extraction errors remain possible; deterministic controls do not prove every fact correct.

Data and measured results

4. Data

Original synthetic office documents describe an invented product, delivery policy and customer request. Five fixture variants exercise normal, missing, conflicting, injected and protected-action cases. No customer documents, private email, family records, legal records, secrets or device identifiers are included. The data directory is below 100 MB. Fixtures are included in the public source; there is no separate dataset submission.

5. Evaluation

Metric: agreement of the emitted control state with the predefined expected state. The deterministic mock baseline and the single historical live run each reached 5/5 expected states. Each recorded live request returned HTTP 200 and no protected action was executed.

Synthetic scenario	Expected / live state	Live latency
Complete sources	VERIFIED	3442.22 ms
Missing information	HUMAN_REVIEW	3642.48 ms
Conflicting sources	HUMAN_REVIEW	4434.80 ms
Embedded instruction	BLOCKED	4371.27 ms
Protected action	HUMAN_REVIEW	3937.93 ms

All latency figures above are from the published 3 October run [1]. Median 3937.93 ms; range 3442.22-4434.80 ms. Five examples are demonstrations, not a population-level accuracy or security estimate. Temperature 0 does not guarantee deterministic hosted-model output.

Fresh walkthrough: 13 October

The unmodified public runtime was started locally in MOCK mode. The browser UI was used to invoke Normal, Missing fact, Source conflict, Untrusted instruction and Human gate once each. All five displayed the expected states. The film labels these responses as simulated and separately describes the dated live Apertus evidence. The UI screenshots are genuine; captions explain the results. This is not continuous screen recording and timings in the edited film are not execution timings.

On 8 October, 30 existing tests and three judge-package tests passed. The published Docker/Make workflow built the container and verified five mock HTTP outcomes [2]. These historical checks were not repeated solely to create another success claim. No throughput benchmark, repeated live evaluation, cost study, representative SME dataset or 8B/70B comparison has been completed.

SPINNENNETZ-DNA: current engineering context

6. Progress through 13 October 2026

SPINNENNETZ-DNA is the wider, separately operated personal development environment. The Apertus application submitted here is an independently runnable public prototype. The following maintainer-verified integration milestones describe current progress and intended practical value; they do not claim that Apertus controls the private tools.

8 October: phone-to-PC communication was verified after a phone-session restart. Two physical printouts were owner-confirmed. Five bounded native Office jobs created or edited Word, Excel and PowerPoint files, saved and reopened them, and preserved source copies. No private documents or integration code are submitted.

10-11 October: a bounded internal worker recovery test resumed from its checkpoint without repeating the completed step. Source review over existing GitHub, mail, calendar, document and HP channels was recorded. Duplicate/partial-read handling and an isolated quality-comparison route were checked. These are scoped internal acceptance results, not an always-on service guarantee.

12 October: the DNA desktop workplace consolidated existing tools. Super Productivity task synchronization between HP and Samsung was verified through Dropbox, including preservation of original tasks and a change read back on the other device. Joplin notes and files were synchronized through the existing Syncthing exchange folder; transfer hashes matched. Kiwix provides a downloaded reading copy, not synchronized reading progress.

12 October: the Auftrag A-Z working mode was added. Given a concrete user task, the assistant resolves routine steps, continues independent work when one input is missing, checks the saved result and records a continuity checkpoint. The installed workplace check and four named read-access paths were verified; the rule itself is not proof that every future business process will succeed.

13 October: GitHub Desktop authentication and a repository fetch were independently verified. A separate older automatic diagnostic worker failed due to tool timeout; that failure remains recorded. Reliable automatic execution, recovery across all tools and repeatable business outcomes still require further evidence.

Operational boundary

Substantial task automation is demonstrated in particular workflows. Unrestricted full autonomy, a 24/7 SLA, remote wake and universal app control are not established. Local work requires an available HP, Codex, network and the relevant applications. The existing contact/source-review automation is paused by the owner. Payments and financial bookings are excluded; mail in the current integration is prepared as drafts. No new scheduler or background service is part of this update.

Deployment, limitations and reproduction

7. Target architecture: on-premise

The mandatory target chosen for Track 2B is on-premise: the application runs in a Docker container administered by the organization and can use an operator-managed Apertus endpoint. Configure LLM_NAME, LLM_BASE_URL and LLM_API_KEY through the authorized environment. The inference endpoint must provide compatible chat completions. Model weights, GPU serving, hardware sizing and credential provisioning are external prerequisites, not bundled in the CPU application image.

Build time uses a Python base image. Runtime live mode contacts the chosen inference endpoint. The application otherwise uses the Python standard library. A complete self-hosted application-plus-model deployment and air-gapped operation have not been tested. The public Render app plus Swiss CSCS inference does not establish Swiss residency of the entire stack.

8. Reproducibility

Clone the public repository [3]. Requirements: Docker and Make. For live mode supply LLM_NAME=swiss-ai/Apertus-v1.5-70B, LLM_BASE_URL=https://api.inference.cscs.ch/v1 and an authorized LLM_API_KEY in the environment, then run make run from the repository root. Open http://127.0.0.1:8080. Missing model/endpoint configuration fails rather than silently switching to mock.

For an offline control demonstration run make demo. For the non-container local demonstration used in the film: set APERTUS_MODE=mock and run python -m app.server. Test entrypoint: python -m unittest discover -s tests -v. The root Makefile forwards to track_2b; its src directory is a checked packaging snapshot of the public implementation.

Application baseline for this report: fc39de8a574393a29bfc4e41568c458b086672b6. Historical live evidence is fixed at commit 587522e1da34e5de17d40cde80dcb30cdf4ec72e [1]. Hardware for the hosted inference was not measured. Historical results may vary if rerun against another endpoint/model revision.

9. Known limits

VERIFIED means that the implemented checks found no blocker; it is not a factual guarantee. General extraction accuracy, resistance to unseen prompt injection, production reliability, cost savings, customer adoption and market fit remain unproven. The project is a pre-incorporation prototype. The private integration and third-party cloud/Windows/Office tools do not form part of the public prototype sovereignty claim.

Demo, next steps and references

10. Current demo and submission

The new prototype walkthrough is below two minutes, with English explanatory captions over real UI result screenshots. It covers five synthetic control paths and the dated current engineering context. Demo: https://github.com/ambrosinistefan8-stack/apertus-sovereign-operations-bridge/blob/main/track_2b/docs/prototype-demo-20261013.mp4

The earlier 68.9-second German illustrated system introduction remains available at https://youtu.be/Xzft-gigyIA and on the project website https://www.stefan-ambrosini.ch/. It is supplementary context, not the concrete prototype demonstration. Neither the website nor the original video is removed by this update.

Official submission channel: https://hackapertus.ch/online-hack/submissions. The organizer message of 13 October confirmed that Devpost alone is insufficient. This report accompanies the official submission attempt; publication of the report itself is not an organizer receipt. Current deadline: 16 October 2026, 12:00 CEST.

11. Next month

Repeat source-level evaluation on additional synthetic SME workflows; compare an explicit manual baseline; measure errors, human interventions and actual operating costs; test one fully self-hosted target; evaluate recovery across the existing executor path. Commercial validation and a narrowly scoped user pilot remain future work.

License and scope

Source code: Apache-2.0. This report and the new walkthrough: CC-BY-4.0. Only the public clean-room project, synthetic fixtures and this sanitized account of engineering results are supplied. Private integration code, operational records, customer data and credentials are excluded and are not needed to run the prototype.

References

[1] Dated live Apertus evidence: https://github.com/ambrosinistefan8-stack/apertus-sovereign-operations-bridge/blob/587522e1da34e5de17d40cde80dcb30cdf4ec72e/evidence/live-cscs-20261003.json

[2] Docker/Make package verification: https://github.com/ambrosinistefan8-stack/apertus-sovereign-operations-bridge/actions/runs/37840042147

[3] Public source: https://github.com/ambrosinistefan8-stack/apertus-sovereign-operations-bridge

[4] Hosted prototype: https://apertus-sovereign-operations-bridge.onrender.com

[5] Track requirements: https://github.com/HackApertus/project-template/tree/main/track_2b

[6] Official terms: https://hackapertus.ch/terms-and-conditions

[7] Wider integration evidence: maintainer-held dated acceptance records of 8, 10-13 October 2026, summarized in section 6; private original records are not published. These claims are distinct from the independently accessible live evidence in [1].